TAO ZHONG

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EDUCATION

Princeton University *08/2023 - 07/2028 (Expected)*
Degree: Ph.D. in MAE (Robotics Track) ***cGPA***: 4.0/4.0
Committee: Christine Allen-Blanchette (*Chair*), Felix Heide, Ryan P. Adams

University of Toronto *09/2018 - 06/2023*
Degree: B.A.Sc. in Engineering Science (with High Honour) ***cGPA***: 3.81/4.0
Major: Robotics Engineering ***Minor***: Artificial Intelligence
Advisor: Animesh Garg

EXPERIENCE

CAB Lab, Princeton University *2023 - Present*
Graduate Research Student, Advisor: Prof. Christine Allen-Blanchette
Topics: physics- and geometry-aware deep learning for scientific discovery [C6, C8, U1, U6. U7], robot learning [C4, C5, C7, U3], and generative foundation models [U2, U4, U5].

People, AI, & Robots Lab, Vector Institute & University of Toronto *2022 - 2023*
Undergraduate Research Student, Advisor: Prof. Animesh Garg
Topics: differentiable physics for dexterous grasp synthesis [C2]

Noah's Ark Lab, Huawei Research Canada *2021 - 2022*
Machine Learning Research Intern, Advisor: Prof. Yang Wang
Topics: OOD generalization via prompt learning [C3] and test-time adaptation [C1] for foundation models

aUToronto, The University of Toronto Self-Driving Car Team *2020 - 2022*
Mapping & Localization Team Lead, Team Advisors: Prof. Tim Barfoot, Prof. Steven Waslander, Prof. Angela Schoellig, Prof. Jonathan Kelly
Topics: semantic mapping and SLAM optimization for autonomous driving.

Shenzhen Institute of Artificial Intelligence and Robotics for Society, CUHK(SZ) *2020*
Visiting Research Student, Advisor: Prof. Huihuan Qian
Topics: web-based testing platform, state estimation, and control for marine robotics

PUBLICATIONS

* denotes equal contribution.

Papers in Submission

[U8] **Tao Zhong**, Kaison Fong, Keqin Wang, Christine Allen-Blanchette. Geometric Algebra for Neural Networks: A Survey. *Preprint, 2026.*

[U7] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Symmetry Descent Neural Operators: From Local Equivariance to Global Consistency. *Under Review, 2026.*

[U6] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Homomorphic Generator Learning: Compositional PDE Neural Operators for Multi-Physics Surrogates. *Under Review, 2026.*

[U5] Yixun Hu, Zhixuan Zhao, **Tao Zhong**, Christine Allen-Blanchette. OrbitRIR: Frame-Averaged Few-Shot RIR Synthesis for 3D Sound Source Localization. *Under Review, 2026.*

[U4] Yixun Hu, Zhixuan Zhao, **Tao Zhong**, Christine Allen-Blanchette. CERPA: Heading-Equivariant Attention for Yaw-Invariant Few-Shot Room-Acoustic Synthesis. *Under Review, 2026.*

[U3] Keqin Wang, **Tao Zhong**, Yulong Yang, Christine Allen-Blanchette. Equivariance Begins at the Source: Characterizing Compatible Sources for Generative Robot Policies. *Under Review, 2026.*

[U2] **Tao Zhong**, Dongzhe Zheng, Christine Allen-Blanchette. HodgeCover: Higher-Order Topological Coverage Drives Compression of Sparse Mixture-of-Experts. *Preprint, 2026*.

[U1] Zhixuan Zhao*, **Tao Zhong***, Yixun Hu, Nathalie P. de Leon, Christine Allen-Blanchette. Neural Fields for NV-Center Inverse Sensing. *Preprint, 2026*.

Refereed Conference Proceedings

[C8] **Tao Zhong**, Yixun Hu, Dongzhe Zheng, Aditya Sood, Christine Allen-Blanchette. Neural Field Thermal Tomography: A Differentiable Physics Framework for Non-Destructive Evaluation. In *Advances in Neural Information Processing Systems (NeurIPS), 2026*.

[C7] Keqin Wang*, **Tao Zhong***, David Chang, Christine Allen-Blanchette. Local-Canonicalization Equivariant Graph Neural Networks for Sample-Efficient and Generalizable Swarm Robot Control. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026*.

[C6] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Topology-Preserving Neural Operator Learning via Hodge Decomposition. In *Proceedings of the International Conference on Machine Learning (ICML), 2026*.

[C5] **Tao Zhong**, Jonah Buchanan, Christine Allen-Blanchette. Grasp2Grasp: Vision-Based Dexterous Grasp Translation via Schrödinger Bridges. In *Advances in Neural Information Processing Systems (NeurIPS), 2025*.

[C4] **Tao Zhong** and Christine Allen-Blanchette. GAGrasp: Geometric Algebra Diffusion for Dexterous Grasping. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2025*.

[C3] Zhixiang Chi*, Li Gu*, **Tao Zhong**, Huan Liu, Yuanhao Yu, Konstantinos N Plataniotis, Yang Wang. Adapting to Distribution Shift by Visual Domain Prompt Generation. In *Proceedings of the International Conference on Learning Representations (ICLR), 2024*.

[C2] Dylan Turpin, **Tao Zhong**, Shutong Zhang, Guanglei Zhu, Eric Heiden, Miles Macklin, Stavros Tsogkas, Sven Dickinson, Animesh Garg. Fast-Grasp'D: Dexterous Multi-finger Grasp Generation Through Differentiable Simulation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2023*.

[C1] **Tao Zhong***, Zhixiang Chi*, Li Gu*, Yang Wang, Yuanhao Yu, Jin Tang. Meta-DMoE: Adapting to Domain Shift by Meta-Distillation from Mixture-of-Experts. In *Advances in Neural Information Processing Systems (NeurIPS), 2022*.

Refereed Non-archival Publications

[W2] Keqin Wang, **Tao Zhong**, Christine Allen-Blanchette. Equivariance Begins at the Source: Quantifying and Characterizing Source Distributions for Generative Robot Policies. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Geometric Representations in Robotics, 2026*.

[W1] **Tao Zhong** and Christine Allen-Blanchette. Geometric Algebra Grasp Diffusion for Dexterous Manipulators. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Equivariant Robotics, 2024*.

Patents

[P1] Zhixiang Chi, Li Gu, **Tao Zhong**, Yuanhao Yu, Yang Wang, Jin Tang. Systems and Methods for Artificial-intelligence Model Training Using Unsupervised Domain Adaptation with Multi-source Meta-distillation. *US Patent Application No. 17/966,568*.

RESEARCH SOFTWARE

NEFI: Neural-Field Inversion with Differentiable Physics [Docs] [Code]
A PyTorch toolkit for inverse imaging and sensing with neural fields and differentiable physics, unifying reconstruction, benchmarking, and diagnostics across modalities without training data.

AWARDS & HONORS

Lambda Research Grant *2026*
Princeton MAE Second Year Departmental Fellowship (2 / 25+) *2024*
Princeton University First Year Fellowship in Natural Sciences and Engineering *2023*
NeurIPS 2022 Scholar Award *2022*

SAE Autodrive Challenge: 1st Place Winner (As a team) *2020, 2021, 2022*
University of Toronto Dean's Honours List (All 8 terms) *2018 - 2023*

INVITED TALKS

Vision-Based Dexterous Grasp Translation via Schrödinger Bridges
GRASP Lab, UPenn, Oct 2025

TEACHING

MAE 206 Introduction to Engineering Dynamics (Undergraduate) *Spring 2026*
Teaching Assistant, Princeton University
MAE 433 Automatic Control Systems (Undergraduate) *Fall 2025, Fall 2026*
Teaching Assistant, Princeton University

MENTORING

Jonah Buchanan (with Christine Allen-Blanchette) *2024 - 2025*
David Chang (with Christine Allen-Blanchette and Keqin Wang) *2024 - 2025*
Kaison Fong (with Christine Allen-Blanchette) *2025*
Zhixuan Zhao (with Christine Allen-Blanchette) *2026*

SERVICE

Conference Refereeing
AAAI Conference on Artificial Intelligence (AAAI) *2027*
International Conference on Machine Learning (ICML) *2026*
Neural Information Processing Systems (NeurIPS) *2025 - 2026*
International Conference on Learning Representations (ICLR) *2025 - 2026*
Annual Learning for Dynamics & Control Conference (L4DC) *2024, 2026*

Journal Refereeing
IEEE Robotics and Automation Letters (RA-L) *2025*
Transactions on Machine Learning Research (TMLR) *2025 - 2026*

SKILLS

Programming Languages: Python, C/C++, MATLAB/Simulink, Bash, SQL, Assembly
Libraries & Tools: PyTorch, NumPy, OpenCV, scikit-learn, ROS, Git, Docker, LaTeX